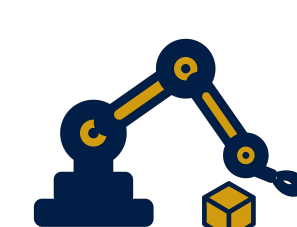


When Simulation Meets Reality, Robots Build the Towers of Hanoi

Sim-to-real pick-and-place on UR5e - from Gazebo debug to live Hanoi Towers
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Overview



Context.

In our lab we utilize a UR5e to develop sim-to-real pick-and-place, demonstrated through the Tower of Hanoi problem.



Challenge.

System engineers use simulation to reduce risk during task development. However, moving from simulation to the real world might create overhead: rewriting task logic, re-tuning motions, and discovering integration bugs on the factory floor.



Contribution.

By constructing high-level primitives such as 'Pick' and 'Place' from low level robot instructions, we set the stage for future task and motion planning activities at CLAIR.

Compose the task once; swap only the deployment underneath.

Execution Pipeline



Tower of Hanoi Pseudocode

```
// pegs: 3 stacks; move top disk only;
// never place large on small
```

```
function Hanoi(n, source, dest, aux):
    if n == 0: return

    Hanoi(n-1, source, aux, dest)

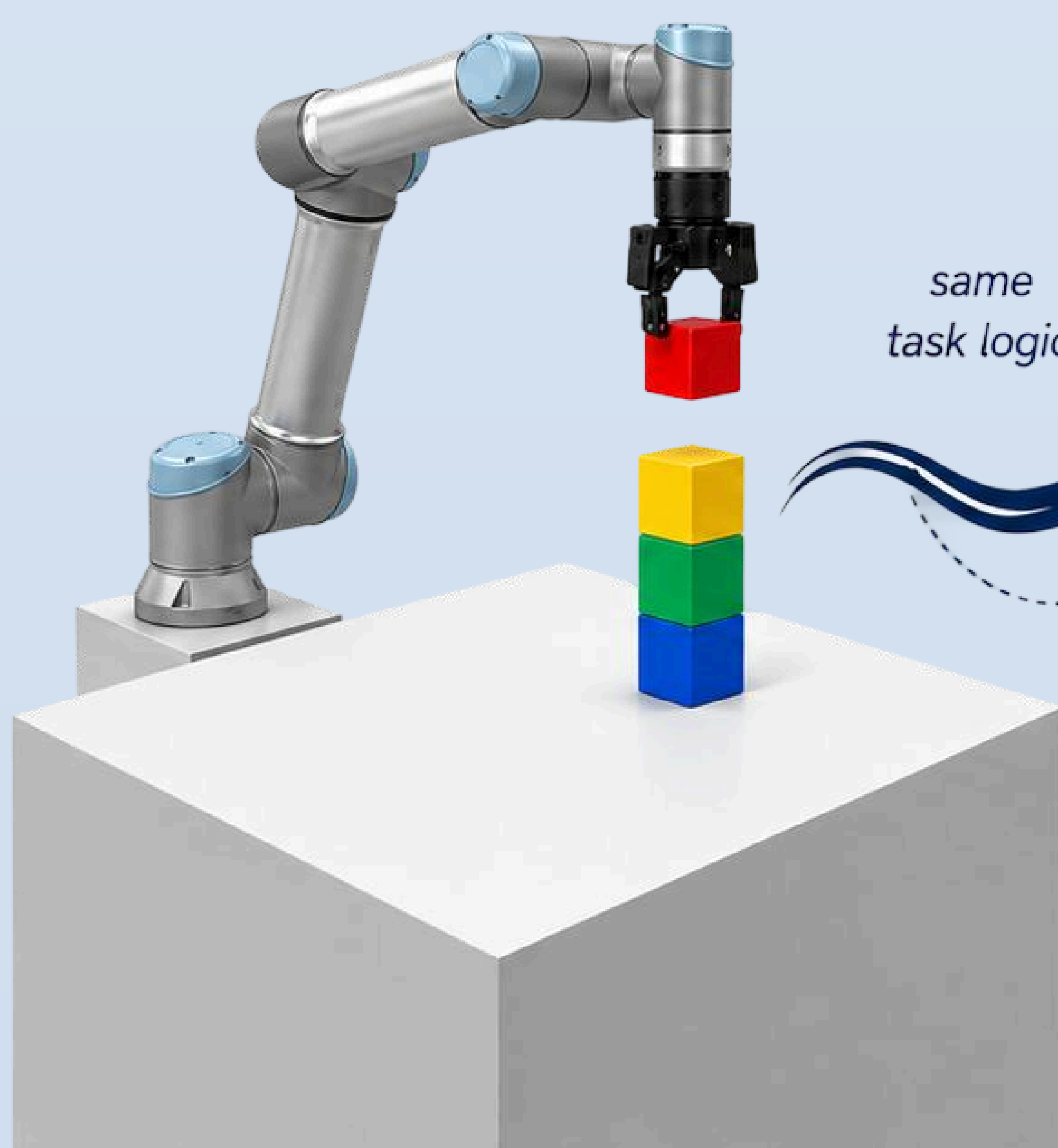
    MoveTopDisk(source → dest)

    Hanoi(n-1, aux, dest, source)
```



Total moves = $2^n - 1$

Gazebo Simulation



same
task logic

ONE YAML PROGRAM

- Pick
- Place
- ExecuteProgram
- SetConstrain

same
task logic

Real UR5e Robot

